
ASSIGNMENT 1

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Question: 6.1

1 Multi-Layer Perceptron with Backpropagation

1.1 Theory

1.1.1 a) Vanishing Gradient Problem

The vanishing gradient problem occurs in deep neural networks when gradients become extremely small during backpropagation, making weight updates negligible for early layers.

Consider a deep network with sigmoid activation:

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

Its derivative is

$$\sigma'(z) = \sigma(z)(1 - \sigma(z))$$

The maximum value of this derivative is

$$\sigma'(z) \leq 0.25$$

During backpropagation, gradients are computed using the chain rule. For a network with L layers:

$$\frac{\partial L}{\partial w^{(1)}} = \frac{\partial L}{\partial a^{(L)}} \prod_{l=2}^L \frac{\partial a^{(l)}}{\partial a^{(l-1)}}$$

Since each term contains the derivative of the sigmoid (0.25), the gradient magnitude shrinks exponentially:

$$|\nabla| \approx (0.25)^L$$

For large L , this becomes extremely small, causing gradients to vanish and preventing early layers from learning effectively.

1.1.2 b) Chain Rule with Randomly Dropped Hidden Units

If hidden units are randomly dropped with probability 0.5, this is similar to the concept of **dropout**.

Let m_i be a binary mask variable:

$$m_i \sim \text{Bernoulli}(0.5)$$

The modified activation becomes

$$a_i = m_i f(z_i)$$

During backpropagation, the chain rule still applies, but gradients are multiplied by the same mask:

$$\frac{\partial L}{\partial z_i} = m_i f'(z_i) \frac{\partial L}{\partial a_i}$$

Thus:

- The chain rule itself does not change.
- Gradients only propagate through the active neurons.
- Neurons that are dropped ($m_i = 0$) receive zero gradient.

This introduces stochasticity in training and helps prevent overfitting while keeping the fundamental backpropagation rule unchanged.